

Derrick Ng

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EDUCATION

Master of Engineering, Artificial Intelligence | GPA: 3.78

Expected Dec 2026

UCLA Samueli School of Engineering | Los Angeles, CA

- *Relevant Coursework:* Artificial Intelligence, Large-Scale Data Mining, Neural Networks & Deep Learning, Natural Language Processing, Automated Reasoning

Bachelor of Arts, Computer Science | GPA: 3.64

May 2025

University of California, Berkeley | Berkeley, CA

SKILLS

Programming & Tools: Python, SQL, NumPy, MySQL, Git, Docker, GCP

Machine Learning: scikit-learn, PyTorch, CNNs, LSTMs, Transformers, Multimodal Fusion, Generative Models

NLP: Hugging Face, LLM Fine-Tuning (SFT, LoRA), BERT/BGE Embeddings, Aspect-Based Sentiment Analysis

Agentic AI: LangGraph, Multi-Agent Orchestration, Pydantic Schema Validation, OWASP-Aligned AI Security

Languages: English, Mandarin Chinese

PROFESSIONAL EXPERIENCE

AI Governance Engineer

June 2026 – Present

iDox.ai (UCLA Samueli Industry Capstone) | Los Angeles, CA

- Building an Azure OpenAI-powered Policy Agent within a LangGraph multi-agent refund workflow, using retrieved policy evidence to generate confidence-scored refund decisions and Pydantic-validated outputs
- Implementing OWASP-aligned AI governance controls at agent handoffs, using audit logs to detect tool-permission violations, semantic drift, and PII leakage, with dashboard alerts for identified risks
- Developing a GCP/MySQL evaluation harness for 20 routine, edge-case, and adversarial refund scenarios, validating workflow reliability through reproducible traces and token-usage tracking

Venture Capital Analyst Intern

May 2025 – Sept 2025

Jinding Capital | Shanghai, China

- Researched Edge-AI SoCs, XR/spatial-computing chips, DPUs/SmartNICs, and sparse accelerators to inform investment thesis development across AI infrastructure
- Conducted technical due diligence on AI semiconductor targets by validating management claims through expert interviews, architecture review, benchmark data, and ecosystem analysis to assess technical defensibility
- Presented VC investment memos on AI semiconductor targets, evaluating commercial viability, competitive positioning, and investment fit for investment committee discussion

Research Assistant

Jun 2024 – August 2025

Co-Design Lab, UC Berkeley | Berkeley, CA

- Built a database of 23,000+ Amazon reviews from 290 sustainability-certified products using automated scraping and a 164-keyword sustainability dictionary across 16 aspects for scalable user-insight extraction
- Developed an NLP workflow using product-feature extraction, aspect-based BERT/GPT sentiment analysis, and LDA topic modeling to structure noisy reviews into design-relevant evidence
- Generated LLM-based sustainable design leads from review-derived context and validated outputs through manual evaluation with correlation analysis for an ASME IDETC-CIE 2025 conference paper

Research Assistant

Jan 2025 – May 2025

Squishy Robotics | Berkeley, CA

- Built a dataset of 540 wildfire sequences from the Corsican Fire Database, pairing cleaned RGB/NIR images with LLM-augmented textual descriptors and computed ground-truth targets for a standardized wildfire-forecasting benchmark
- Implemented a late-fusion multimodal forecasting prototype with EfficientNet-B0, BERT, Transformer fusion, and LSTM regression, reducing validation error by 84% while predicting decomposed fire-progression attributes for explainable wildfire forecasting

Web Development Engineer Intern

Sept 2023 – Dec 2023

AoSheng Global Inc. | El Monte, CA

- Built a full-stack education-consulting website using Next.js/React, designing student-facing navigation and SEO-oriented page architecture to support marketing-driven traffic acquisition
- Designed AWS-backed enterprise data workflows to consolidate fragmented consulting operations into a standardized system for student lifecycle management

PUBLICATIONS

First Author: “Machine Learning-Aided Supply Chain Analysis of Waste Management Systems,” *Sustainability*, 17(19), 8848, 2025

Co-author: “Data-Driven Sustainable Design Opportunities from Automated User Insights,” *ASME IDETC-CIE*, DETC2025-169019, 2025